CAPPING WITH THE FIRST CHERN CLASS OF AN INVERTIBLE SHEAF FACTORS THROUGH RATIONAL EQUIVALENCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a coproduct and proper pushforward reduction to a single rational function on an integral space, followed by an exchange of the invertible sheaf for the structure sheaf through commutativity of caps

Editorial note. The label lemma-factors names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Rational equivalence by locally finite sums of divisors of rational functions on integral closed subspaces of one dimension higher, and the definition of the cap operation with a first Chern class
- The compatibility of the cap operation with proper pushforward, which allows a computation on a coproduct to descend
- The commutativity of the caps with two invertible sheaves on an integral space, and the additivity of the cap operation in the invertible sheaf

Source reference. Original reference 02SO: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 2569–2569. Prerequisite pointer invoked at proof line(s) 3340; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02RW: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 1918–1918. Prerequisite pointer invoked at proof line(s) 3341; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02SP: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 2614–2614. Prerequisite pointer invoked at proof line(s) 3368; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02TH: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 3304–3304. Prerequisite pointer invoked at proof

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line(s) 3363; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02SU: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 2933–2933. Prerequisite pointer invoked at proof line(s) 3351; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02QL: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 264–264. Prerequisite pointer invoked at proof line(s) 3328; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 02QL let X/B be good. Let $\mathcal{L}$ be invertible on X. The operation $\alpha \mapsto c_1(\mathcal{L}) \cap \alpha$ factors through rational equivalence to give an operation*

$$c_1(\mathcal{L}) \cap - : \mathrm{CH}_{k+1}(X) \to \mathrm{CH}_k(X)$$

Proof. Let $\alpha \in Z_{k+1}(X)$, and $\alpha \sim_{rat} 0$. We have to show that $c_1(\mathcal{L}) \cap \alpha$ as defined in Definition 02SO is zero. By Definition 02RW there exists a locally finite family $\{W_j\}$ of integral closed subspaces with $\dim_\delta(W_j) = k + 2$ and rational functions $f_j \in R(W_j)^*$ such that

$$\alpha = \sum (i_j)_* \mathrm{div}_{W_j}(f_j)$$

Note that $p : \coprod W_j \to X$ is a proper morphism, and hence $\alpha = p_*\alpha'$ where $\alpha' \in Z_{k+1}(\coprod W_j)$ is the sum of the principal divisors $\mathrm{div}_{W_j}(f_j)$. By Lemma 02SU we have $c_1(\mathcal{L}) \cap \alpha = p_*(c_1(p^*\mathcal{L}) \cap \alpha')$. Hence it suffices to show that each $c_1(\mathcal{L}|_{W_j}) \cap \mathrm{div}_{W_j}(f_j)$ is zero. In other words we may assume that X is integral and $\alpha = \mathrm{div}_X(f)$ for some $f \in R(X)^*$.

Assume X is integral and $\alpha = \mathrm{div}_X(f)$ for some $f \in R(X)^*$. We can think of f as a regular meromorphic section of the invertible sheaf $\mathcal{N} = \mathcal{O}_X$. Choose a meromorphic section s of $\mathcal{L}$ and denote $\beta = \mathrm{div}_{\mathcal{L}}(s)$. By Lemma 02TH we conclude that

$$c_1(\mathcal{L}) \cap \alpha = c_1(\mathcal{O}_X) \cap \beta.$$

However, by Lemma 02SP we see that the right hand side is zero in $\mathrm{CH}_k(X)$ as desired. □